

Acute Mechanical and Metabolic Responses to Different Resistance Training Protocols With Equated Volume Load

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Purpose: To investigate the effect of different resistance training protocols with equated volume load on acute mechanical and metabolic responses. **Methods:** In a randomized order, 18 men performed 8 different training protocols in the bench press exercise consisting of (sets, repetitions, intensity, and interset recoveries) 3 × 16, 40% 1-repetition maximum (1RM), 2 and 5 minutes; 6 × 8, 40% 1RM, 2 and 5 minutes; 3 × 8, 80% 1RM, 2 and 5 minutes; and 6 × 4, 80% 1RM, 2 and 5 minutes. Volume load was equalized between protocols (1920 arbitrary units). Velocity loss and effort index were calculated during the session. Movement velocity against the 60% 1RM and blood lactate concentration pre–post exercise were used to assess the mechanical and metabolic responses, respectively. **Results:** Resistance training protocols performed with heavy load (80% 1RM) resulted in a lower ($P < .05$) total number of repetitions (effect size = -2.44) and volume load (effect size = -1.79) than the scheduled ones when longer set configurations and shorter rest periods were used in the same protocol (ie, higher-training-density protocols). Protocols including a higher number of repetitions per set and shorter rest times induced higher velocity loss, effort index, and lactate concentrations than the rest of the protocols. **Conclusions:** Our results suggest that resistance training protocols with similar volume load but different training variables (ie, intensity, number of sets and repetitions, rest between sets) produce different responses. Implementing a lower number of repetitions per set and longer rest intervals is recommended to reduce the intrasession and postsession fatigue.

Keywords: interset recovery, intensity, lactate, velocity loss, fatigue

Muscular strength plays an important role in general (eg, jumping and sprinting) and specific sports skills.¹ Accordingly, conducting regular resistance training (RT) is a method commonly used to improve athletic performance.² Manipulation of RT variables (ie, load, number of repetitions and sets, and rest period lengths between sets, among others)³ characterizes the workload and, consequently, physiological adaptations to RT.⁴ Therefore, quantification of the workload in RT is necessary to allow coaches to modulate training stressors and the recovery–adaptation process, which provides greater potential to maximize performance gains.⁵ This is important to consider as RT is usually part of the total training carried out in different sports disciplines, sometimes being conducted concurrently with other more sport-specific sessions (eg, technical or tactical training sessions).

To monitor the workload, the volume load is generally calculated.⁵ One of the most-used equations to calculate the volume load is the product of the number of repetitions completed in an exercise (ie, total training volume) by the percentage of 1-repetition maximum (%1RM).⁶ This equation is represented as follows: volume load (in arbitrary units) = number of sets × number of repetitions × %1RM. Volume load is related to measures of internal workload and physiological response during RT at various intensities⁷ and is typically used to compare RT protocols with different configurations.⁸ However, this

approach has several drawbacks. First, a similar volume load can be obtained with different set configurations; for example, 3 sets of 8 repetitions with 80% 1RM have a similar volume load to 6 sets of 8 repetitions with 40% 1RM (volume load = 1920 arbitrary unit). These RT configurations, despite having the same volume load, have important differences in 2 of the most important variables that define the workload: volume (ie, 24 vs 48 repetitions) and intensity (80% vs 40% of 1RM).

Previous studies have shown that matched-intensity and matched-volume protocols (using different set configurations but similar volume loads) may result in different acute mechanical and metabolic responses^{9–11} and, as a result, in specific long-term adaptations.¹² In this regard, it has been observed that training configurations of 3 sets of 10 repetitions with 10RM load (3 × 10 [10]) resulted in a significantly higher acute decline in physical performance (eg, countermovement jump height or velocity with 75% 1RM) and a higher increase in markers of fatigue (eg, blood lactate, creatine kinase, or ammonia concentrations) when compared with completing the same total volume (ie, 6 × 5 [10]) with a different set configuration.^{10,11} Similarly, 5 sets of 10 repetitions with 10RM load in the leg press exercise resulted in greater decreases in power output and depletion of muscle energy metabolites stores (ie, muscle phosphocreatine and adenosine triphosphate), along with higher levels of muscle fatigue indicators (ie, lactate and inosine monophosphate), than 10 sets of 5 repetitions with the same load.⁹ These results indicate that despite using the same load and total volume (which results in the same volume load), longer set configurations (ie, more repetitions per set) are associated with higher mechanical and metabolic markers of fatigue, anaerobic glycolysis, and decreases in performance, which can potentially impact the recovery and adaptation processes.^{9–11}

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Another potential drawback of using volume load to quantify RT workloads is that this formula does not consider the rest interval length between or even within sets. Rest interval length influences both mechanical and metabolic acute effects of training.¹³ In this regard, a 2-minute rest interval between sets resulted in a significant decrease in the number of repetitions completed in the bench press (BP) and squat exercises with light and heavy loads,^{14–16} whereas a 5-minute rest resulted in higher volume completed and provided an adequate recovery to maintain an 8 to 12 repetitions range.^{14,15} In addition, rest intervals of 3 to 5 minutes may be safer and more reliable for maximal strength assessment^{17,18} and for better maintenance of high mean velocities¹⁹ and power production²⁰ over multiple sets and repetitions. These latter authors also observed that resting for less than 2 minutes resulted in a significant elevation in blood lactate concentration, whereas longer rest intervals only induced slight increases.²⁰ On the other hand, it has recently been proposed that abbreviating interset rest times and including shorter but more frequent rest intervals (ie, rest redistribution set structure) is an effective method for maintaining mechanical performance, mitigating fatigue, and reducing disruption to the energy balance and metabolic response in the muscle.²¹

Therefore, the purpose of this study was to investigate the acute mechanical and metabolic response to 8 RT protocols with the same scheduled volume load but different loading (ie, 40% and 80% 1RM), total volume configurations (ie, sets $\times$ repetitions), and interset rest intervals. We hypothesized that RT protocols with lower numbers of repetitions per set and longer rest times would elicit lower mechanical (velocity loss) and metabolic (blood lactate concentration) fatigue than the opposite configuration.

Methods

Subjects

Eighteen resistance-trained men (age 28.3 [5.2] y; height 1.73 [0.05] m; body mass 80.5 [10.5] kg) with at least 2 years of RT experience in the BP exercise (range 2–8 y; 1RM strength for the BP exercise: 88.4 [16.6] kg, and 1.10 [0.17] normalized per kg of body mass) volunteered to participate in this study. All participants were injury free and were fully informed about the procedures, potential risks, and benefits of the study, and they all signed a written informed consent prior to the tests. The study was conducted in accordance with the Declaration of Helsinki II and was approved by the research ethics committee of the local university.

Design

A randomized and cross-sectional experimental design was undertaken to examine the acute mechanical and metabolic responses to 8 RT sessions with different loading, set configurations, and rest time distribution but the same total volume load between protocols (see Table 1, for details). A total of 8 RT protocols were designed as follows: (1) 3 sets of 16 repetitions with 40% 1RM and 2-minute interset recovery (L40V3 $\times$ 16R2), (2) 3 sets of 16 repetitions with 40% 1RM and 5-minute interset recovery (L40V3 $\times$ 16R5), (3) 6 sets of 8 repetitions with 40% 1RM and 2-minute interset recovery (L40V6 $\times$ 8R2), (4) 6 sets of 8 repetitions with 40% 1RM and 5-minute interset recovery (L40V6 $\times$ 8R5), (5) 3 sets of 8 repetitions with 80% 1RM and 2-minute interset recovery (L80V3 $\times$ 8R2), (6) 3 sets of 8 repetitions with 80% 1RM and 5-minute interset recovery (L80V3 $\times$ 8R5), (7) 6 sets of 4 repetitions with 80% 1RM and 2-minute interset recovery (L80V6 $\times$ 4R2), and (8) 6 sets of 4

repetitions with 80% 1RM and 5-minute interset recovery (L80V6 $\times$ 4R5). Following familiarization with the BP execution technique (ie, lifting at maximal velocity and pausing before the concentric phase) and recording the individual grip width to standardize the range of movement throughout the experiment, subjects were randomly assigned to perform one of the 8 RT protocols separated by a period of 3 to 6 days. Intensities were determined from the individual load–velocity relationship. Participants were asked to abstain from any strenuous physical activity for at least 2 days before each session. All sessions took place at a research laboratory under the direct supervision of a researcher. All sessions were performed on a Smith machine with no counterweight mechanism (Matrix Fitness, G1-FW161) at the same time of day for each subject and under similar environmental conditions (20 °C and 60% humidity, approximately).

Methodology

Progressive Loading Test. A detailed description of this test has been provided elsewhere.²² All repetitions were recorded at 1000 Hz with a linear velocity transducer (T-Force System Ergotech), the reliability of which has been previously reported.²³ The velocity measures used in this study corresponded to the mean velocity of the propulsive phase of each repetition (MPV).²⁴ Only the best repetition (ie, highest MPV) with each load was considered for subsequent analysis. The individualized load–velocity relationships were obtained from these tests.

Resistance Exercise Protocols. Resting lactate concentrations were obtained before the warm-up. Then, subjects performed a standardized warm-up consisting of 5 minutes of joint mobilization exercises followed by 3 sets of 8, 6, and 4 repetitions with 20 kg and 40% and 50% of 1RM, respectively. After the warm-up, 3 repetitions with 60% 1RM load were performed, and the average velocity of these 3 repetitions was taken as a baseline strength measure. The absolute loads (in kilograms) were individually adjusted to ensure that the corresponding MPV matched ($\pm 0.03 \text{ m}\cdot\text{s}^{-1}$) the prescribed %1RM for each session. We used a range of $0.03 \text{ m}\cdot\text{s}^{-1}$ as it has been shown that this value is the smallest detectable change in MPV when using the T-Force System.²³ The battery of tests was repeated as follows: blood lactate (60-s post), and 3 repetitions against the 60% of 1RM load (90-s Post), to obtain the acute postexercise values (see Figure 1, for details). The fastest MPV attained in each RT protocol, average MPV attained during the session (A-MPV), average velocity loss magnitude induced within the set (VL, defined as: $100 \times [\text{MPV of the last repetition} - \text{MPV of the fastest repetition}] / \text{MPV of the fastest repetition}$), and effort index (EI; defined as: fastest MPV attained in each RT protocol $\times$ average %VL²⁵) were calculated to analyze the differences between RT protocols. In addition, the training sessions were split into 3 parts to compare the changes in performance with a matched-volume load among RT protocols. Thus, the session was split as follows: 1/3: data reported from the first part of the session (ie, in protocols using 40% 1RM, consisting of 48 repetitions, data reported from the first 16 repetitions; in protocols using 80% 1RM, consisting of 24 repetitions, from the first 8 repetitions); 2/3: data reported from the second part of the session (ie, from the second 16 and 8 repetitions); and 3/3: data reported from the last part of the session (ie, from the last 16 and 8 repetitions).

Measurement of Fatigue and Metabolic Stress. The percentage of VL attained against the 60% 1RM load was used as an indicator

Table 1 Descriptive Characteristics of the Scheduled and Actually Performed Resistance Training Protocols

	L40V3 × 16R2	L40V3 × 16R5	L40V6 × 8R2	L40V6 × 8R5	L80V3 × 8R2	L80V3 × 8R5	L80V6 × 4R2	L80V6 × 4R5
Scheduled								
Set × repetitions	3 × 16	3 × 16	6 × 8	6 × 8	3 × 8	3 × 8	6 × 4	6 × 4
Total repetitions	48	48	48	48	24	24	24	24
Intensity, %1RM	40	40	40	40	80	80	80	80
Rest time, min	2	5	2	5	2	5	2	5
Total rest time, min	4	10	10	25	4	10	10	25
Total volume load (set × rep × %1RM)	1920	1920	1920	1920	1920	1920	1920	1920
Actually performed								
Fastest MPV, m·s ⁻¹	1.03 (0.07) (0.92–1.16)	1.04 (0.07) (0.91–1.18)	1.04 (0.07) (0.93–1.19)	1.01 (0.07) (0.90–1.15)	0.42 (0.06) (0.32–0.50)	0.44 (0.04) (0.36–0.51)	0.43 (0.04) (0.37–0.48)	0.45 (0.05) (0.36–0.53)
Intensity, %1RM	39.8 (1.6) (37.7–43.3)	39.6 (1.2) (37.8–42.6)	39.7 (1.1) (37.3–41.1)	39.3 (1.1) (36.8–41.8)	79.8 (1.9) (76.6–83.0)	79.4 (2.0) (75.6–84.5)	79.3 (1.7) (77.0–82.8)	78.3 (2.0) (74.5–82.6)
Total repetitions	47.7 (1.4) (42–48)	48.0 (0.0) (48–48)	48.0 (0.0) (48–48)	48.0 (0.0) (48–48)	16.3 (4.7) ^{f,g,h} (7–24)	21.2 (3.3) ^h (14–24)	23.3 (1.8) (18–24)	24.0 (0.0) (24–24)
Total volume load (set × rep × %1RM)	1896.3 (100.9) ^{e,f} (1846.1–1946.2)	1903.0 (56.7) ^{e,f} (1874.8–1931.2)	1907.8 (52.1) ^{e,f} (1881.8–1933.7)	1885.4 (55.1) ^{e,f} (1858.0–1912.8)	1294.6 (361.7) ^{f,g,h} (1114.7–1474.5)	1680.3 (232.4) ^h (1564.7–1795.8)	1845.0 (135.5) (1777.6–1912.3)	1879.9 (48.1) (1856.0–1903.9)

Abbreviations: 1RM, 1-repetition maximum; MPV, mean propulsive velocity; rep, repetition. Note: Data are presented as mean (SD) and range (between brackets). Protocols: L40V3 × 16R2: 3 × 16, 40% 1RM, 2-minute rest. L40V3 × 16R5: 3 × 16, 40% 1RM, 5-minute rest. L40V6 × 8R2: 6 × 8, 40% 1RM, 2-minute rest. L40V6 × 8R5: 6 × 8, 40% 1RM, 5-minute rest. L80V3 × 8R2: 3 × 8, 80% 1RM, 2-minute rest. L80V3 × 8R5: 3 × 8, 80% 1RM, 5-minute rest. L80V6 × 4R2: 6 × 4, 80% 1RM, 2-minute rest. L80V6 × 4R5: 6 × 4, 80% of 1RM, 5-minute rest. Parameters: Total reps performed; total volume load: calculated as sets × reps × %1RM; fastest MPV: the fastest repetition (usually the first) measured in each resistance training protocol (this value represents the average intensity, %1RM).

Statistically significant differences ($P < .05$) between the different resistance training protocols are as follows: ^aL40V3 × 16R5, ^bL40V3 × 16R5, ^cL40V6 × 8R2, ^dL40V6 × 8R5, ^eL80V3 × 8R2, ^fL80V3 × 8R5, ^gL80V6 × 4R2, and ^hL80V6 × 4R5.

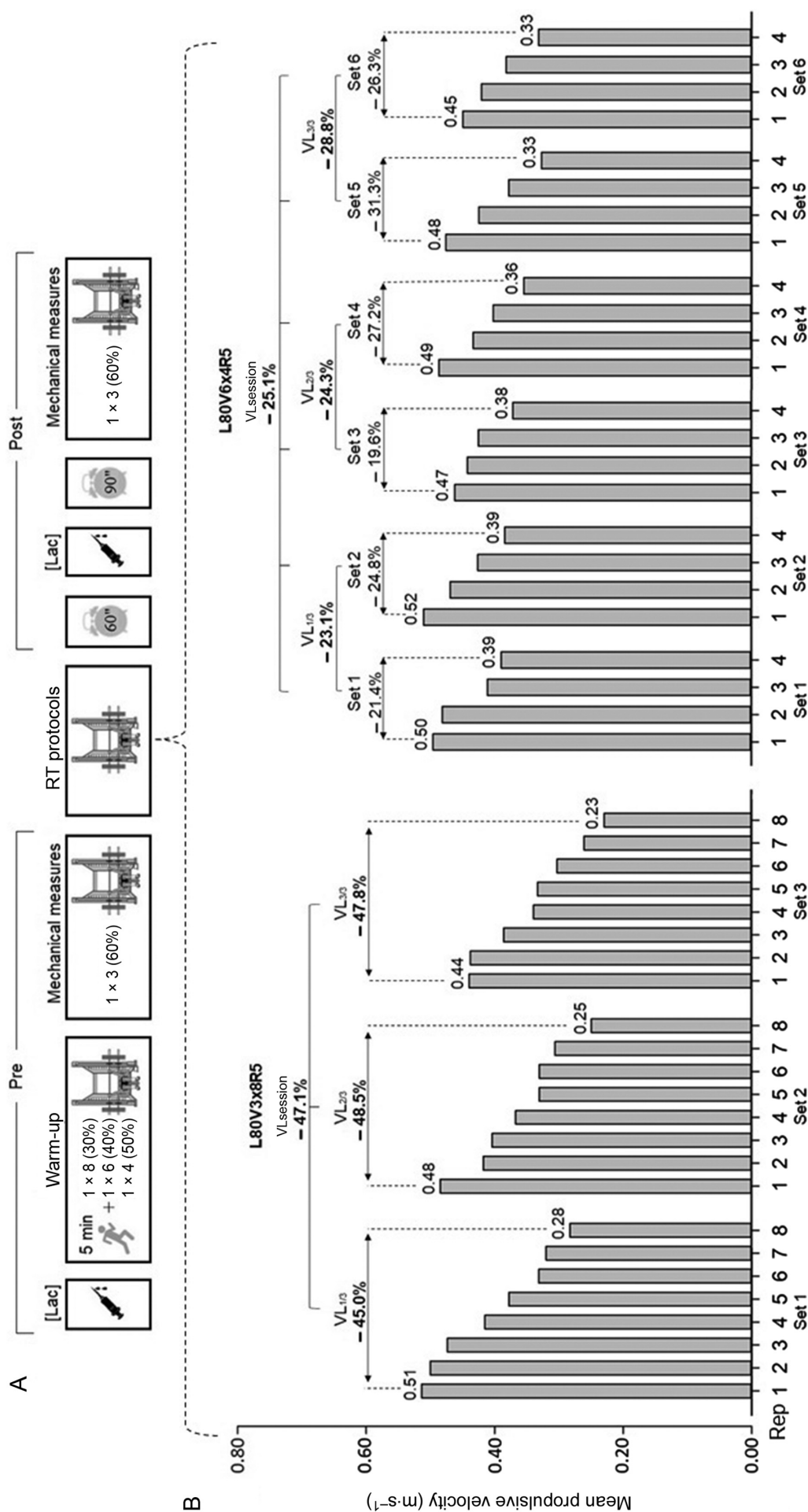


Figure 1 — (A) Schematic representation of the experimental design. The order of the RT protocols was randomized. (B) Example of the quantification of intrasession velocity loss for 2 RT protocols with different set distribution. Data obtained from a representative subject. Subscripts: session: data reported from the entire session; 1/3: data reported from the first third of the session; 2/3: data reported from the second third of the session; 3/3: data reported from the last third of the session. [Lac] indicates blood lactate concentrations; Rep, repetitions; RT, resistance training; VL, magnitude of velocity loss.

of the mechanical fatigue induced by each protocol. Blood lactate concentration was used as an indicator of the metabolic response induced by each protocol. A whole blood sample of about 5 μL , obtained from the fingertip, was used for the quantification of lactate concentration, which was measured with a portable lactate analyzer (Lactate Pro 2; ARKRAY) at 2 different time points: before and 1 minute after the training protocol. The reliability and accuracy of Lactate Pro 2 analyzer have been previously established with coefficients of variation of 7.6%, 3.5%, and 2.7% for lactate concentrations of 1, 4, and 12 $\text{mmol}\cdot\text{L}^{-1}$, respectively.²⁶

Statistical Analysis

Data are presented as mean (SD) values. A one-way repeated-measures analysis of variance was used on the fastest MPV attained in each RT protocol, intensity (%1RM), total repetitions, and total volume load of the session to compare the RT characteristics of the 8 RT protocols. In addition, this analysis was used to analyze the differences in velocity-derived variables (ie, MPV, VL, and EI) between the RT protocols during the entire training session. A 2-factor (protocol $\times$ time and protocol $\times$ sets) repeated-measures analysis of variance was calculated for mechanical fatigue, lactate concentration, and velocity-derived variables (from the 3 sections into which all the training sessions were divided). Bonferroni post hoc tests were used when the interaction was significant. Correlations between lactate, VL, and EI were reported using Pearson product-moment correlation coefficients (r). In addition, effect sizes (ESs) and their confidence intervals were calculated using Hedges g on the pooled SD. Significance was accepted at $P \leq .05$. All statistical analyses were performed using SPSS software (version 20.0).

Results

Table 1 provides the descriptions of the scheduled and performed RT protocols for each loading condition. The actual average intensity was 39.6% (1.3%) and 79.2% (1.9%) without significant differences between protocols with the same intensity. The average total repetitions performed during protocols using 40% 1RM was 47.9 (0.7) repetitions (48 scheduled repetitions) without significant differences between them. However, the average total repetitions performed during protocols with 80% 1RM were 21.2 (4.2) repetitions (24 scheduled repetitions) with significant differences ($P < .01$) between protocols (Table 1). The average volume load performed was 1786.5 (257.7) arbitrary units (1920 scheduled volume load) wherein L80V3 $\times$ 8R2 and L80V3 $\times$ 8R5 resulted in a significantly lower volume load ($P < .05$; ES range -3.31 to -1.22) than the other RT protocols.

Table 2 shows the descriptive characteristics of the velocity-derived parameters in each RT protocol. Significant differences ($P < .001$) were observed in A-MPV between protocols. Lower A-MPV values were observed for protocols with higher numbers of repetitions per set (ie, longer set configurations). In addition, a shorter rest time between sets accentuated the reduction in A-MPV (eg, $P = .27$, ES = 0.64 [0.32–0.97] for L40V3 $\times$ 16). Similarly, higher ($P < .001$) VL and EI were observed for protocols that had longer set configurations and shorter total rest times.

Analyzing the intraprotocol response, significant decreases ($P < .05$) in A-MPV were observed for L40V3 $\times$ 16R2, L80V3 $\times$ 8R2, and L80V6 $\times$ 4R2. Similarly, significant increases ($P < .05$) in VL and EI were observed during L40V3 $\times$ 16R2 (ES = 1.53 [0.83–2.24];

ES = 1.55 [0.84–2.26], respectively), whereas the rest of the protocols remained unchanged. Figure 2 shows the evolution of the MPV attained in each repetition for each RT protocol.

Table 3 shows the acute mechanical and metabolic response to each RT protocol. Significant differences ($P < .001$) were observed for both variables between premeasurement and postmeasurement for all protocols. Blood lactate concentrations and MPV at 60% of 1RM significantly differed between protocols ($P < .001$ and $P = .03$, respectively). Post hoc comparisons revealed higher blood lactate concentrations and lower MPV with 60% of 1RM for longer set configurations and shorter total rest times.

Significant ($P < .001$) relationships were observed between EI and VL with 60% 1RM, between EI and blood lactate concentration, and between blood lactate concentration and VL with 60% 1RM (Figure 3).

Discussion

The present study described the acute mechanical and metabolic responses following 8 different RT protocols in which the scheduled volume load was equated between them, but loading conditions, rest time length, and set configuration were manipulated. Our main results indicate that manipulating intensity, rest time, and set configuration induces different mechanical and metabolic responses, despite using a matched-volume load for all RT protocols. Furthermore, as the training protocol approached muscle failure, the mechanical and metabolic response increased. The descriptive characteristics of the velocity-derived parameters revealed lower A-MPV and higher VL and EI for RT protocols with longer set configurations and shorter rest periods. In addition, shorter set configurations (ie, more frequent rest times) and longer rest times produced a reduced mechanical and metabolic response, assessed by the pre–post changes in the MPV attained with 60% of 1RM load, and the blood lactate concentration. Therefore, intensity, set configurations (ie, shorter sets), and rest time and its distribution (ie, longer and more frequent intersets) seem to modulate the stimuli induced during RT, despite a main variable traditionally employed to monitor workload during RT, such as volume load, being equated for all protocols.

In the present study, we observed that RT protocols with longer set configurations, shorter rest periods, and higher intensity (ie, L80V3 $\times$ 8R2) resulted in a significantly lower total number of repetitions completed (ie, total volume) and volume load than the other RT protocols analyzed. It is worth noting that, regardless of the 5-minute rest, the number of repetitions performed decreased throughout sets in the protocols consisting of 3 $\times$ 8 with 80% 1RM. During these RT protocols, 17 and 10 of 18 subjects reached muscle failure in L80V3 $\times$ 8R2 and L80V3 $\times$ 8R5, respectively. This fact is explained because, with the 80% 1RM, approximately 8 maximum repetitions can be performed (coefficient of variation [CV] = 14%).²² This means that the level of effort of these sets was maximum (up to task failure) and that the fatigue induced by previous sets prevented subjects from performing scheduled repetitions. As a result, they could not achieve the scheduled volume load. This is also supported by the high VL observed for the 80% 1RM protocols using long set configurations (56.6% and 58.4% for L80V3 $\times$ 8R2 and L80V3 $\times$ 8R5, respectively). However, when long set configurations were used with 40% 1RM, the VL values were lower (35.4% and 28.2% for L40V3 $\times$ 16R2 and L40V3 $\times$ 16R5, respectively). In the BP exercise, a 50% VL means that the set is conducted to task failure, whereas a 25% VL suggests that the

Table 2 Descriptive Characteristics of Velocity-Derived Parameters for Each Resistance Training Protocol

	L40V3 × 16R2	L40V3 × 16R5	L40V6 × 8R2	L40V6 × 8R5	L80V3 × 8R2	L80V3 × 8R5	L80V6 × 4R2	L80V6 × 4R5
A-MPV _{session} , m·s ⁻¹	0.83 (0.08) ^{c,d}	0.89 (0.08) ^d	0.94 (0.08)	0.96 (0.07)	0.27 (0.04) ^{g,h}	0.30 (0.03) ^h	0.33 (0.05)	0.36 (0.05)
A-MPV _{1/3} , m·s ⁻¹	0.90 (0.06) ^{c,d}	0.90 (0.07) ^{c,d}	0.96 (0.07)	0.96 (0.06)	0.31 (0.05) ^h	0.31 (0.04) ^h	0.35 (0.05)	0.37 (0.05)
A-MPV _{2/3} , m·s ⁻¹	0.84 (0.09) ^{b-d}	0.89 (0.07) ^{c,d}	0.95 (0.08)	0.96 (0.07)	0.26 (0.04) ^{*,f-h}	0.30 (0.03) ^h	0.32 (0.06)	0.36 (0.05)
A-MPV _{3/3} , m·s ⁻¹	0.77 (0.12) ^{*,b-d}	0.87 (0.09) ^{c,d}	0.93 (0.08)	0.95 (0.07)	0.22 (0.04) ^{*,f-h}	0.28 (0.04) ^h	0.30 (0.06) [*]	0.34 (0.06)
VL _{session} , %	35.4 (9.3) ^{c-f}	28.2 (6.6) ^{c-f}	13.7 (4.5) ^{e-h}	12.0 (3.1) ^{e-h}	56.6 (7.8) ^{g,h}	58.4 (8.2) ^{g,h}	34.9 (8.4)	30.6 (7.8)
VL _{1/3} , %	27.0 (6.0) ^{c-f}	26.7 (6.3) ^{c-f}	13.0 (4.3) ^{e-h}	11.8 (2.9) ^{e-h}	57.6 (11.4) ^{g,h}	58.0 (8.2) ^{g,h}	32.1 (9.2)	27.9 (7.4)
VL _{2/3} , %	34.6 (9.6) ^{c-f}	28.7 (6.6) ^{c-f}	13.7 (4.9) ^{e-h}	12.4 (3.8) ^{e-h}	56.7 (14.2) ^{g,h}	57.9 (8.4) ^{g,h}	37.5 (10.4)	29.9 (8.3)
VL _{3/3} , %	44.6 (15.1) ^{*,†c,d,f}	29.2 (8.6) ^{c-f}	14.3 (4.9) ^{d-g}	11.7 (3.6) ^{e-h}	55.6 (12.1) ^{g,h}	59.2 (13.2) ^{g,h}	35.2 (14.1)	33.9 (10.0)
EI _{session} , AU	36.4 (9.4) ^{b-h}	29.1 (6.2) ^{c,d,g,h}	14.1 (4.5) ^{e,f}	12.4 (3.3) ^{e,f}	24.1 (5.6) ^{g,h}	25.4 (4.0) ^{g,h}	14.4 (3.6)	12.6 (2.8)
EI _{1/3} , AU	27.9 (6.8) ^{c,d,g,h}	27.7 (6.5) ^{c,d,g,h}	13.4 (4.4) ^{e,f}	12.3 (3.1) ^{e,f}	24.3 (6.1) ^{g,h}	25.2 (3.8) ^{g,h}	13.6 (3.3)	12.3 (2.9)
EI _{2/3} , AU	35.6 (9.8) ^{c-h}	29.6 (6.4) ^{c,d,g,h}	14.1 (4.8) ^{e,f}	12.8 (3.9) ^{e,f}	24.4 (8.0) ^{g,h}	25.2 (3.9) ^{g,h}	15.9 (3.8)	13.2 (3.2)
EI _{3/3} , AU	45.7 (14.7) ^{*,†b-h}	30.1 (8.0) ^{c,d,g,h}	14.8 (4.8) ^{d,e,f}	12.2 (3.7) ^{e,f}	23.6 (6.4) ^{g,h}	25.8 (6.2) ^{g,h}	15.2 (6.1)	14.9 (3.4)

Abbreviations: 1RM, 1-repetition maximum; AU, arbitrary unit; A-MPV, average mean propulsive velocity; EI, effort index; VL, magnitude of velocity loss. Note: Data are presented as mean (SD). The data were compared within and between relative intensities except A-MPV, which was compared only within the same relative intensities as different intensities imply distinct velocities. Protocols: L40V3 × 16R2: 3 × 16, 40% 1RM, 2-minute rest. L40V3 × 16R5: 3 × 16, 40% 1RM, 5-minute rest. L40V6 × 8R2: 6 × 8, 40% 1RM, 2-minute rest. L40V6 × 8R5: 6 × 8, 40% 1RM, 5-minute rest. L80V3 × 8R2: 3 × 8, 80% 1RM, 2-minute rest. L80V3 × 8R5: 3 × 8, 80% 1RM, 5-minute rest. L80V6 × 4R2: 6 × 4, 80% 1RM, 2-minute rest. L80V6 × 4R5: 6 × 4, 80% of 1RM, 5-minute rest. Parameters: A-MPV reported from the corresponding repetitions expressed as subscript; VL calculated as: 100 · ([velocity of the last repetition – velocity of the fastest repetition]/velocity of the fastest repetition), experienced for the corresponding repetitions expressed as subscript; EI, calculated as: the fastest velocity · VL, for the corresponding repetitions expressed as subscript.

Subscripts: session: data reported from the entire session; 1/3: data reported from the first third of the session; 2/3: data reported from the second third of the session; 3/3: data reported from the last third of the session.

Statistically significant differences ($P < .05$) between the different protocols are as follows: ^aL40V3 × 16R5, ^bL40V3 × 16R5, ^cL40V6 × 8R2, ^dL40V6 × 8R5, ^eL80V3 × 8R2, ^fL80V3 × 8R5, ^gL80V6 × 4R2, and ^hL80V6 × 4R5.

Statistically significant differences with 1/3: * $P < .05$. Statistically significant differences with 2/3: † $P < .05$.

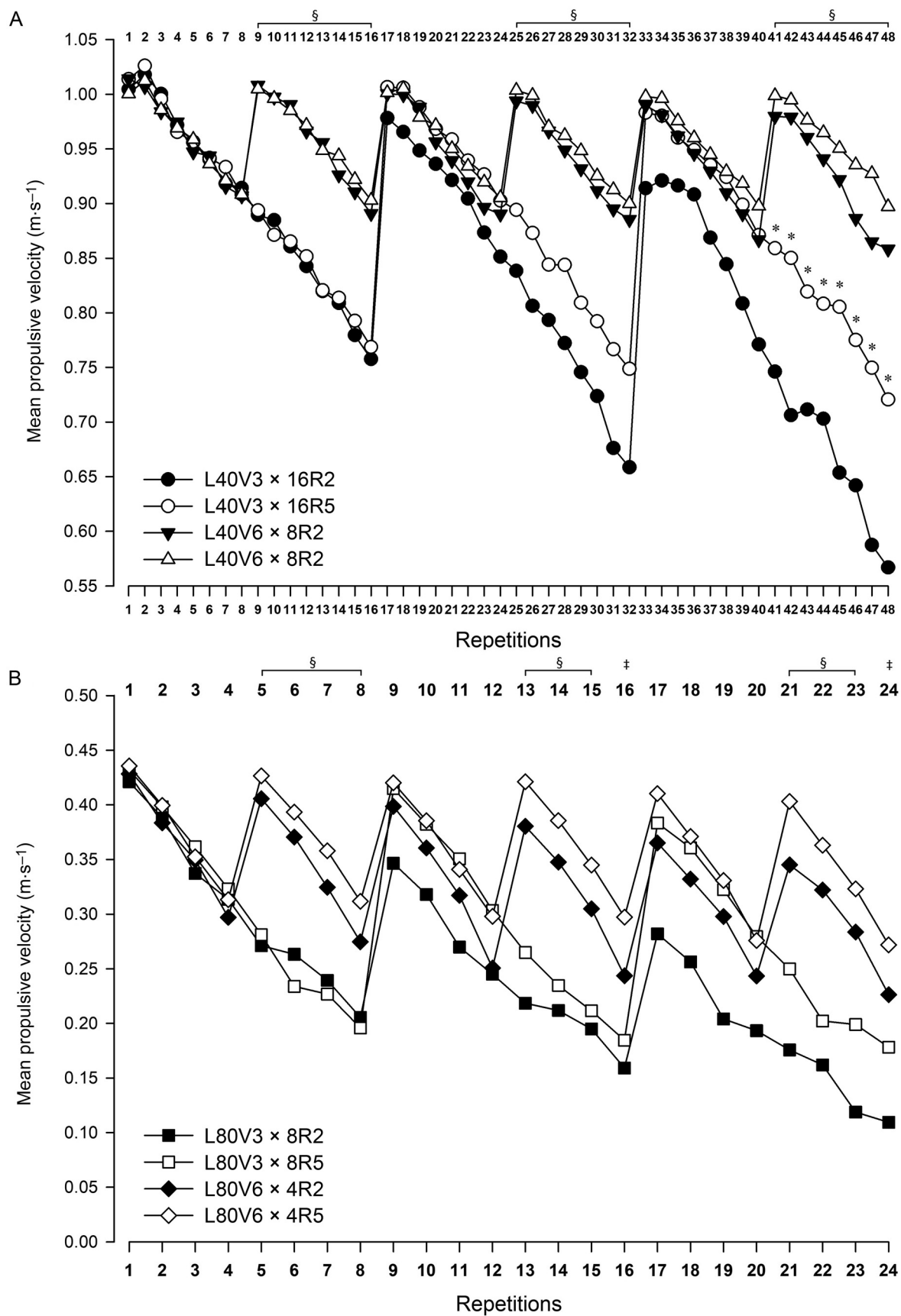


Figure 2 — Evolution of mean propulsive velocity attained at each repetition for each resistance training protocol performed against the 40% of 1RM load (A) and the 80% of 1RM load (B). L40V3 $\times$ 16R2: 3 $\times$ 16, 40% 1RM, 2-minute rest. L40V3 $\times$ 16R5: 3 $\times$ 16, 40% 1RM, 5-minute rest. L40V6 $\times$ 8R2: 6 $\times$ 8, 40% 1RM, 2-minute rest. L40V6 $\times$ 8R5: 6 $\times$ 8, 40% 1RM, 5-minute rest. L80V3 $\times$ 8R2: 3 $\times$ 8, 80% 1RM, 2-minute rest. L80V3 $\times$ 8R5: 3 $\times$ 8, 80% 1RM, 5-minute rest. L80V6 $\times$ 4R2: 6 $\times$ 4, 80% 1RM, 2-minute rest. L80V6 $\times$ 4R5: 6 $\times$ 4, 80% of 1RM, 5-minute rest. 1RM indicates 1-repetition maximum. *Significant differences between RT protocols with the same set configurations but different interval rests. §Significant differences between RT protocols with different set configurations. †Significant differences between L80V3 $\times$ 8R5 and L80V6 $\times$ 4R5. $P < .05$.

Table 3 Acute Mechanical and Metabolic Response to Each Resistance Training Protocol

	L40V3 × 16R2	L40V3 × 16R5	L40V6 × 8R2	L40V6 × 8R5	L80V3 × 8R2	L80V3 × 8R5	L80V6 × 4R2	L80V6 × 4R5
Lactate, mmol·L ⁻¹								
Pre	1.4 (0.2)	1.5 (0.3)	1.3 (0.2)	1.6 (0.4)	1.4 (0.2)	1.3 (0.2)	1.4 (0.5)	1.4 (0.4)
Post	8.5 (1.6)*,d,g,h	8.0 (2.5)*,d	6.8 (2.6)*,d	4.7 (1.0)*,e-g	8.6 (2.8)*,g,h	8.6 (2.7)*,g	6.5 (1.8)*	5.9 (2.5)*
MPV 60% of 1RM load, m·s ⁻¹								
Pre	0.68 (0.05)	0.69 (0.04)	0.69 (0.06)	0.68 (0.06)	0.68 (0.06)	0.69 (0.04)	0.68 (0.06)	0.68 (0.06)
Post	0.50 (0.09)*,b-d,h	0.57 (0.09)*,d,e	0.63 (0.07)*,e,f	0.64 (0.07)*,e-h	0.47 (0.08)*,g,h	0.51 (0.10)*,h	0.56 (0.10)*	0.60 (0.07)*
Velocity loss, %	27.2 (9.9) ^{c,d,h}	17.7 (10.0) ^{d,e}	8.2 (6.7) ^{e-g}	6.5 (4.4) ^{e-g}	29.8 (9.8) ^{g,h}	26.6 (12.6) ^h	18.0 (9.9)	12.1 (6.2)

Abbreviations: 1RM, 1-repetition maximum; Lactate, blood lactate concentration; MPV 60% 1RM, average of the mean propulsive velocity attained during the 3 repetitions with the 60% of 1RM; Pre, baseline value; Post, value obtained after performing the corresponding protocol. Note: Data are presented as mean (SD). Protocols: L40V3 × 16R2: 3 × 16, 40% 1RM, 2-minute rest. L40V3 × 16R5: 3 × 16, 40% 1RM, 5-minute rest. L40V6 × 8R2: 6 × 8, 40% 1RM, 2-minute rest. L40V6 × 8R5: 6 × 8, 40% 1RM, 5-minute rest. L80V3 × 8R2: 3 × 8, 80% 1RM, 2-minute rest. L80V3 × 8R5: 3 × 8, 80% 1RM, 5-minute rest. L80V6 × 4R2: 6 × 4, 80% 1RM, 2-minute rest. L80V6 × 4R5: 6 × 4, 80% of 1RM, 5-minute rest.

Statistically significant differences (*P* < .05) between the different resistance exercise sessions are as follows: ^aL40V3 × 16R5, ^bL40V3 × 16R5, ^cL40V6 × 8R2, ^dL40V6 × 8R5, ^eL80V3 × 8R2, ^fL80V3 × 8R5, ^gL80V6 × 4R2, and ^hL80V6 × 4R5.

Statistically significant differences with pre: **P* < .001.

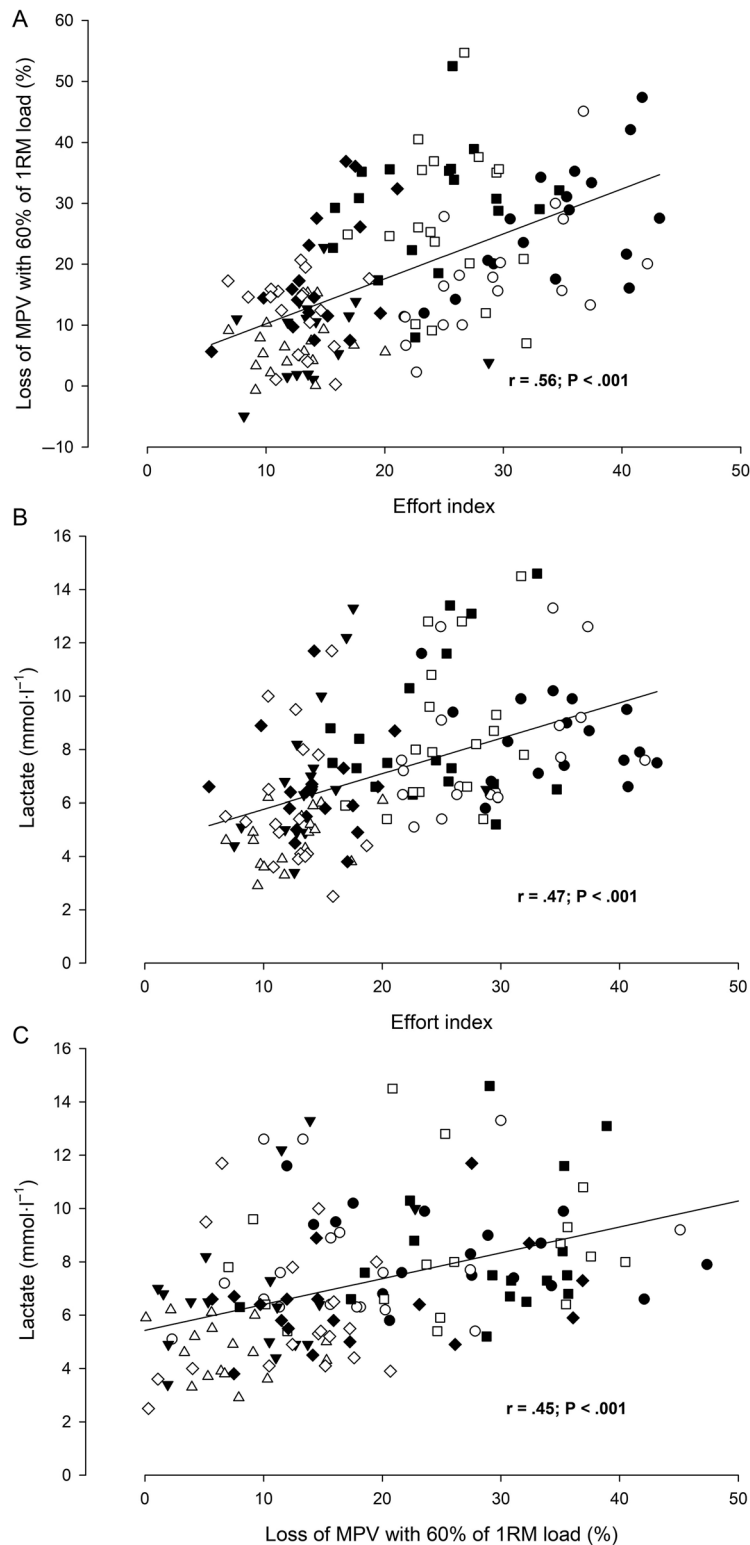


Figure 3 — Relationships between EI and loss of MPV pre-post exercise against the 60% of 1RM load (A) and postexercise lactate concentration (B), and relationship between loss of MPV pre-post exercise against the 60% of 1RM load and postexercise lactate concentration (C). L40V3 × 16R2: 3 × 16, 40% 1RM, 2-minute rest. L40V3 × 16R5: 3 × 16, 40% 1RM, 5-minute rest. L40V6 × 8R2: 6 × 8, 40% 1RM, 2-minute rest. L40V6 × 8R5: 6 × 8, 40% 1RM, 5-minute rest. L80V3 × 8R2: 3 × 8, 80% 1RM, 2-minute rest. L80V3 × 8R5: 3 × 8, 80% 1RM, 5-minute rest. L80V6 × 4R2: 6 × 4, 80% 1RM, 2-minute rest. L80V6 × 4R5: 6 × 4, 80% of 1RM, 5-minute rest. Different symbols are used to differentiate between the different RT protocol analyzed: L40V3 × 16R2 (black circle), L40V3 × 16R5 (white circle), L40V6 × 8R2 (black triangle down), L40V6 × 8R5 (white triangle up), L80V3 × 8R2 (black square), L80V3 × 8R5 (white square), L80V6 × 4R2 (black diamond), and L80V6 × 4R5 (white diamond). 1RM indicates 1-repetition maximum; EI, effort index; MPV, mean propulsive velocity.

athlete has performed half-maximal repetitions.²⁷ Previous studies have observed a significant decrease in the number of repetitions and work (ie, repetitions $\times$ weight) when it is performed with a maximum number of repetitions for a given load (ie, 15 repetitions with a 15RM load) or to volitional exhaustion (leading to failure) with different rest time lengths.^{14,16} These last authors concluded that a 5-minute rest was not enough to fully recover between sets, and recovery was significantly worse when rest times were less than 3 minutes. However, in our study, when RT protocols were performed not leading to failure, no significant differences were observed in total volume and volume load, regardless of using light or heavy loads and 2- or 5-minute rests (Table 1). Therefore, as RT is performed closer to muscle failure, longer interset rest should be required to maintain repetitions and volume load scheduled. Based on our results, it seems that more than a 5-minute rest is needed to maintain the total repetitions during RT sessions conducted to muscular failure, whereas when approximately a half-maximal number of repetitions per set is performed, it seems that a 2-minute rest is enough.

Regarding velocity-derived parameters, lower A-MPV and higher VL were observed for the longer set configurations (Table 2). When the intrasession response was analyzed, a significant decrease in A-MPV (and increase in VL) was found throughout the sets, specifically in those RT protocols with longer set configurations and shorter rest intervals (ie, L40V3 $\times$ 16R2 and L80V3 $\times$ 8R2). Our findings support previous research reporting that proximity to failure impairs lifting kinematics, resulting in slower velocity and lower force production.²⁸ In this regard, Gorostiaga et al⁹ reported greater power impairments along with higher depletion of intramuscular adenosine triphosphate and phosphocreatine stores after 5 $\times$ 10 versus 10 $\times$ 5 with 85% 1RM with 2-minute interset rests in the leg press exercise. In addition to set configuration, shorter rest times exacerbate fatigue accumulation during the training session (Table 2). This can be expected because it is known that longer rest intervals increase the recovery of bioenergetic factors such as adenosine triphosphate and phosphocreatine.²⁹ Our results agree with previous studies, which recommended at least 3 minutes of rest time for the maintenance of high velocities and higher power production during RT sets.^{19,20}

The acute effects of the RT protocols showed a similar trend to the aforementioned during the training sessions as longer set configurations and shorter rest times (ie, higher training density protocols) produced the largest mechanical and metabolic responses. Specifically, a significantly lower postexercise performance was observed for L40V3 $\times$ 16R2 compared with the same protocol with a 5-minute rest. As we mentioned earlier, a 2-minute rest may not be enough for adenosine triphosphate and phosphocreatine levels to recover after performing a high number of repetitions (16 repetitions per set) as it has been proposed that 3 minutes is necessary for total recovery of the high-energy phosphate system.²⁹ This explanation is supported by the higher lactate concentrations observed in set configurations with a 2-minute rest (ES: 0.43; 95% confidence interval, -0.23 to 1.09 ; $P > .05$). Supporting this observation, Abdessemed et al²⁰ reported higher blood lactate concentrations during an RT protocol (10 $\times$ 6 repetitions at 70%1RM in BP exercise) using interset rests of 1 minute compared with 3 and 5 minutes. When RT protocols were performed with a high level of effort (ie, close to task failure), similar decreases in postexercise performance were observed between training configurations with 2 and 5 minutes of rest (L80V3 $\times$ 8R2 vs L80V3 $\times$ 8R5). Again, these findings support

the proposition that when volume load is equated in RT protocols using different loads, those protocols using heavier loads induce higher fatigue as even 5 minutes of rest was not enough to alleviate the acute mechanical and metabolic response to these protocols. This is also supported by the high VL observed for these protocols ($> 50\%$), which results in high EI. EI is considered an accurate expression of neuromuscular fatigue induced during RT,²⁵ and it has been proposed as a tool to compare the acute and chronic effects of different RT protocols.³⁰ Based on EI data, the most demanding RT protocols were (Table 2): (1) L40V3 $\times$ 16R2, (2) L40V3 $\times$ 16R5, (3) L80V3 $\times$ 8R2, and (4) L80V3 $\times$ 8R5 (these last 2 were about the same level). Accordingly, it could be expected that mechanical and metabolic responses would follow that order. However, following postexercise performance and blood lactate response, the most demanding protocols were (Table 3): (1) L80V3 $\times$ 8R2, (2) L80V3 $\times$ 8R5, (3) L40V3 $\times$ 16R2 (these 3 were about the same level), and (4) L40V3 $\times$ 16R5. This can be explained because when a set to failure has been performed, the VL and the EI are already maximal for such loads. However, if more sets are performed, the VL and the EI will be the same or even lower (because the fastest velocity of the set is going to decrease), but the fatigue experienced during the training session is higher. Moreover, significant relationships were observed between EI and changes in pre–post exercise performance ($r = .56$) and between EI and postexercise lactate concentration ($r = .47$) (Figure 2). Therefore, RT protocols that resulted in higher EI showed higher postsession fatigue. However, the correlations were lower than those previously reported ($r = .95$ – $.98$).³⁰ This could be because Rodríguez-Rosell et al³⁰ examined relationships using only the average value of each RT protocol analyzed, which removes interindividual variability and considerably increases the magnitude of the relationship observed between 2 variables. Thus, although EI may be an appropriate parameter to quantify training fatigue experienced with different loads during RT, further research is required to address these questions.

One limitation of the study is that all participants were “beginner” to “intermediate” resistance-trained individuals, according to Santos et al.³¹ Therefore, the acute response of another training status (ie, “advanced” or “highly advanced”) remains unexplored. Future studies should also explore the acute response using other exercises (eg, squat or pull-up), intensities (eg, 30% and 60% 1RM), level of effort (ie, proximity to failure), and populations (eg, women). Finally, given the small size of the sample ($n = 18$), the correlation data shown in the present study should be taken with caution.

Practical Applications

RT sessions with shorter set configurations and longer rest interval lengths are recommended to reduce intrasession and postsession fatigue. Our results suggest that when RT is performed close to or to muscular failure, more than 5 minutes of rest may be needed to reduce the fatigue induced during RT. However, during RT sessions performed well below muscular failure (approximately half-maximal numbers of repetitions per set), 2 minutes of rest may be enough to maintain high movement velocities and low VL and EI during the successive sets. When comparing protocols with different loads but similar volume load, those protocols using heavier loads induced higher levels of fatigue, mainly because they were closer to task failure. In addition to monitoring the workload, VL and EI seem to provide more accurate and objective information about the acute fatigue induced during RT than volume load.

Conclusions

The present data show that shorter set configurations (ie, more frequent rest times) and longer rest times resulted in the highest mean velocities, the lowest values of VL and EI, and a reduced mechanical and metabolic postsession response. Of note is that those protocols performed close to or to failure resulted in higher levels of fatigue, regardless of the intensity and interset rest time used.

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